

ZIMA, Russian Federation

WMO: 306030

Lat: 53.933N

Lon: 102.050E

Elev: 458

StdP: 95.94

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.3	-33.9	-39.6	0.1	-37.1	-36.2	0.1	-33.7	7.8	-10.7	6.3	-12.2	0.1	0	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	29.4	18.4	27.6	17.7	25.9	17.1	20.0	26.2	19.1	25.2	18.3	24.0	2.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.7	22.2	17.0	12.9	21.7	16.1	12.1	21.0	59.8	26.1	56.6	25.2	53.9	23.8	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
7.3	6.0	4.8	-38.3	32.7	3.9	1.9	-41.1	34.0	-43.4	35.1	-45.6	36.2	-48.5	37.6		
			-38.3	22.2	3.8	1.9	-41.0	23.6	-43.2	24.8	-45.3	25.9	-48.0	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-20.5	-16.6	-7.4	2.8	9.7	16.3	18.9	16.2	9.0	0.9	-10.3	-17.9
	DBStd	14.75	6.97	6.96	6.56	4.77	4.56	3.84	2.84	3.41	4.33	5.09	7.12	7.36
	HDD10.0	4344	946	745	541	219	63	2	0	2	69	284	608	864
	HDD18.3	6712	1204	978	799	465	270	83	28	83	280	542	858	1122
	CDD10.0	761	0	0	0	4	53	192	276	194	41	1	0	0
	CDD18.3	86	0	0	0	0	2	22	45	17	1	0	0	0
	CDH23.3	1077	0	0	0	2	72	331	457	200	16	0	0	0
	CDH26.7	261	0	0	0	0	15	91	112	41	2	0	0	0
Wind	WSAvg	1.6	0.9	1.1	1.7	2.4	2.4	2.0	1.7	1.7	1.7	1.6	1.4	1.0
Precipitation	PrecAvg	387	13	10	9	18	32	58	74	72	47	22	19	19
	PrecMax	579	28	19	25	40	79	143	169	156	123	83	43	50
	PrecMin	211	3	2	1	6	6	19	13	25	19	2	6	3
	PrecStd	89	6	5	6	9	20	29	38	32	23	15	10	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.4	1.3	9.4	20.9	28.6	31.8	31.8	30.0	25.7	17.8	5.8	-1.0
		MCWB	-3.5	-2.0	3.4	9.9	14.9	18.3	20.5	18.7	15.5	9.4	2.2	-2.9
	2%	DB	-5.4	-1.7	6.3	17.3	25.3	29.2	29.5	27.6	22.3	13.6	2.8	-3.7
		MCWB	-6.3	-3.8	1.6	8.2	13.4	17.6	19.3	18.2	13.9	7.4	0.4	-4.7
	5%	DB	-8.2	-4.3	4.1	14.2	22.4	27.1	27.6	25.6	19.5	10.7	1.0	-5.9
		MCWB	-9.1	-6.0	0.4	6.6	11.9	16.9	18.6	17.4	12.6	5.9	-0.5	-6.7
	10%	DB	-10.7	-6.8	2.0	11.3	19.4	24.9	25.9	23.5	16.8	8.2	-0.6	-7.9
		MCWB	-11.3	-8.1	-1.1	5.2	10.4	16.1	18.2	16.6	11.2	4.7	-1.8	-8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.3	-1.4	3.9	10.7	15.8	20.0	22.0	20.5	16.5	10.1	2.9	-2.6
		MCDB	-2.4	1.0	8.3	19.1	26.4	26.6	28.8	27.2	23.5	16.2	5.3	-1.3
	2%	WB	-6.2	-3.8	2.1	8.8	14.1	18.9	20.6	19.1	14.7	8.1	0.7	-4.8
		MCDB	-5.4	-1.7	6.2	16.1	23.6	26.2	26.8	25.4	20.5	12.5	2.1	-3.8
	5%	WB	-9.0	-5.9	0.6	7.1	12.6	17.9	19.7	18.2	13.2	6.4	-0.4	-6.7
		MCDB	-8.2	-4.3	3.7	13.5	20.8	25.3	25.5	24.1	18.4	10.1	1.1	-6.0
	10%	WB	-11.3	-8.0	-0.9	5.4	11.1	16.9	18.9	17.2	11.8	4.9	-1.9	-8.5
		MCDB	-10.7	-6.7	1.9	11.0	18.5	23.1	24.5	22.2	15.8	7.8	-0.5	-8.0
Mean Daily Temperature Range	5% DB	MDBR	9.3	12.2	12.7	11.0	12.9	12.2	10.9	11.0	11.1	9.9	8.7	8.3
		MCDBR	9.6	12.7	13.3	16.6	18.8	16.7	14.9	15.4	15.9	14.6	9.1	8.4
		MCWBR	8.8	10.6	9.2	9.1	8.8	6.8	6.2	6.9	7.8	8.5	7.1	7.6
	5% WB	MCDBR	9.4	12.3	12.1	15.3	16.8	13.8	12.0	13.2	14.2	13.5	8.7	8.3
		MCWBR	8.7	10.3	8.5	8.9	8.3	6.2	5.7	6.4	7.3	8.2	7.0	7.6
Clear-Sky Solar Irradiance	taub	0.279	0.307	0.340	0.385	0.435	0.427	0.423	0.391	0.347	0.327	0.304	0.280	
	taud	2.041	2.028	2.043	2.177	2.088	2.220	2.264	2.336	2.417	2.344	2.154	2.016	
	Ebn at Noon	644	757	820	838	811	823	819	826	824	748	614	547	
	Edh at Noon	72	103	128	129	150	133	125	110	89	76	65	61	
All-Sky Solar Radiation	RadAvg	1.07	2.15	3.74	4.85	5.38	5.95	5.38	4.58	3.24	1.95	1.04	0.74	
	RadStd	0.07	0.12	0.18	0.32	0.44	0.35	0.48	0.28	0.28	0.15	0.09	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page